A First Course in PROBABILITY

EIGHTH EDITION



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Now, using that

$$1 + \frac{q_n}{p_n - p_i} = \frac{p_n - p_i + q_n}{p_n - p_i} = \frac{q_i}{p_n - p_i}$$

the preceding gives

$$\begin{aligned} P\{S_n = k\} &= \sum_{i=1}^{n-1} p_i q_i^{k-1} \prod_{i \neq j \leq n} \frac{p_j}{p_j - p_i} + p_n q_n^{k-1} \prod_{j<n} \frac{p_j}{p_j - p_n} \\ &= \sum_{i=1}^{n} p_i q_i^{k-1} \prod_{j \neq i} \frac{p_j}{p_j - p_i} \end{aligned}$$

and the proof by induction is complete. ■

6.4 CONDITIONAL DISTRIBUTIONS: DISCRETE CASE

Recall that, for any two events E and F, the conditional probability of E given F is defined, provided that $P(F) > 0$, by

$$P(E|F) = \frac{P(EF)}{P(F)}$$

Hence, if X and Y are discrete random variables, it is natural to define the conditional probability mass function of X given that $Y = y$, by

$$\begin{aligned} p_{X|Y}(x|y) &= P\{X = x|Y = y\} \\ &= \frac{P\{X = x, Y = y\}}{P\{Y = y\}} \\ &= \frac{p(x, y)}{p_Y(y)} \end{aligned}$$

for all values of y such that $p_Y(y) > 0$. Similarly, the conditional probability distribution function of X given that $Y = y$ is defined, for all y such that $p_Y(y) > 0$, by

$$\begin{aligned} F_{X|Y}(x|y) &= P\{X \leq x|Y = y\} \\ &= \sum_{a \leq x} p_{X|Y}(a|y) \end{aligned}$$

In other words, the definitions are exactly the same as in the unconditional case, except that everything is now conditional on the event that $Y = y$. If X is independent of Y, then the conditional mass function and the distribution function are the same as the respective unconditional ones. This follows because if X is independent of Y, then

$$\begin{aligned} p_{X|Y}(x|y) &= P\{X = x|Y = y\} \\ &= \frac{P\{X = x, Y = y\}}{P\{Y = y\}} \\ &= \frac{P\{X = x\}P\{Y = y\}}{P\{Y = y\}} \\ &= P\{X = x\} \end{aligned}$$

EXAMPLE 4a

Suppose that $p(x, y)$, the joint probability mass function of X and Y, is given by

$$p(0,0) = .4 \quad p(0,1) = .2 \quad p(1,0) = .1 \quad p(1,1) = .3$$

Calculate the conditional probability mass function of X given that $Y = 1$.

Solution. We first note that

$$p_Y(1) = \sum_x p(x,1) = p(0,1) + p(1,1) = .5$$

Hence,

$$p_{X|Y}(0|1) = \frac{p(0,1)}{p_Y(1)} = \frac{2}{5}$$

and

$$p_{X|Y}(1|1) = \frac{p(1,1)}{p_Y(1)} = \frac{3}{5}$$ ■

EXAMPLE 4b

If X and Y are independent Poisson random variables with respective parameters λ_1 and λ_2, calculate the conditional distribution of X given that $X + Y = n$.

Solution. We calculate the conditional probability mass function of X given that $X + Y = n$ as follows:

$$\begin{aligned} P\{X = k|X + Y = n\} &= \frac{P\{X = k, X + Y = n\}}{P\{X + Y = n\}} \\ &= \frac{P\{X = k, Y = n - k\}}{P\{X + Y = n\}} \\ &= \frac{P\{X = k\}P\{Y = n - k\}}{P\{X + Y = n\}} \end{aligned}$$

where the last equality follows from the assumed independence of X and Y. Recalling (Example 3e) that $X + Y$ has a Poisson distribution with parameter $\lambda_1 + \lambda_2$, we see that the preceding equals

$$\begin{aligned} P\{X = k|X + Y = n\} &= \frac{e^{-\lambda_1}\lambda_1^k}{k!}\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}\left[\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}\right]^{-1} \\ &= \frac{n!}{(n-k)!\,k!}\frac{\lambda_1^k\lambda_2^{n-k}}{(\lambda_1+\lambda_2)^n} \\ &= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k} \end{aligned}$$

In other words, the conditional distribution of X given that $X + Y = n$ is the binomial distribution with parameters n and $\lambda_1/(\lambda_1 + \lambda_2)$. ■

We can also talk about joint conditional distributions, as is indicated in the next two examples.

EXAMPLE 4c

Consider the multinomial distribution with joint probability mass function

$$P\{X_i = n_i, i = 1, \ldots, k\} = \frac{n!}{n_1! \cdots n_k!} p_1^{n_1} \cdots p_k^{n_k}, \quad n_i \geq 0, \sum_{i=1}^{k} n_i = n$$

Such a mass function results when n independent trials are performed, with each trial resulting in outcome i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. The random variables X_i, $i = 1, \ldots, k$, represent, respectively, the number of trials that result in outcome i, $i = 1, \ldots, k$. Suppose we are given that n_j of the trials resulted in outcome j, for $j = r + 1, \ldots, k$, where $\sum_{j=r+1}^{k} n_j = m \leq n$. Then, because each of the other $n - m$ trials must have resulted in one of the trials $1, \ldots, r$, it would seem that the conditional distribution of $X_1, \ldots, X_r$ is the multinomial distribution on $n - m$ trials with respective trial outcome probabilities

$$P\{\text{outcome } i | \text{outcome is not any of } r + 1, \ldots, k\} = \frac{p_i}{F_r}, \; i = 1, \ldots, r$$

where $F_r = \sum_{i=1}^{r} p_i$ is the probability that a trial results in one of the outcomes $1, \ldots, r$.

Solution. To verify this intuition, let $n_1, \ldots, n_r$, be such that $\sum_{i=1}^{r} n_i = n - m$. Then

$$\begin{aligned} P\{X_1 = n_1, \ldots, X_r = n_r | X_{r+1} = n_{r+1}, \ldots X_k = n_k\} & \\ &= \frac{P\{X_1 = n_1, \ldots, X_k = n_k\}}{P\{X_{r+1} = n_{r+1}, \ldots X_k = n_k\}} \\ &= \frac{\frac{n!}{n_1! \cdots n_k!} p_1^{n_1} \cdots p_r^{n_r} p_{r+1}^{n_{r+1}} \cdots p_k^{n_k}}{\frac{n!}{(n-m)! n_{r+1}! \cdots n_k!} F_r^{n-m} p_{r+1}^{n_{r+1}} \cdots p_k^{n_k}} \end{aligned}$$

where the probability in the denominator was obtained by regarding outcomes $1, \ldots, r$ as a single outcome having probability F_r, thus showing that the probability is a multinomial probability on n trials with outcome probabilities $F_r, p_{r+1}, \ldots, p_k$. Because $\sum_{i=1}^{r} n_i = n - m$, the preceding can be written as

$$\begin{aligned} P\{X_1 = n_1, \ldots, X_r = n_r | X_{r+1} = n_{r+1}, \ldots X_k = n_k\} & \\ &= \frac{(n - m)!}{n_1! \cdots n_r!} \left(\frac{p_1}{F_r}\right)^{n_1} \cdots \left(\frac{p_r}{F_r}\right)^{n_r} \end{aligned}$$

and our intuition is upheld. ■

EXAMPLE 4d

Consider n independent trials, with each trial being a success with probability p. Given a total of k successes, show that all possible orderings of the k successes and $n - k$ failures are equally likely.

Solution. We want to show that, given a total of k successes, each of the $\binom{n}{k}$ possible orderings of k successes and $n - k$ failures is equally likely. Let X denote the number of successes, and consider any ordering of k successes and $n - k$ failures, say, $\mathbf{o} = (s, s, f, f, \ldots, f)$. Then

$$\begin{aligned} P(\mathbf{o}|X = k) &= \frac{P(\mathbf{o}, X = k)}{P(X = k)} \\ &= \frac{P(\mathbf{o})}{P(X = k)} \\ &= \frac{p^k(1 - p)^{n-k}}{\binom{n}{k}p^k(1 - p)^{n-k}} \\ &= \frac{1}{\binom{n}{k}} \end{aligned}$$

■

6.5 CONDITIONAL DISTRIBUTIONS: CONTINUOUS CASE

If X and Y have a joint probability density function $f(x, y)$, then the conditional probability density function of X given that $Y = y$ is defined, for all values of y such that $f_Y(y) > 0$, by

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

To motivate this definition, multiply the left-hand side by dx and the right-hand side by $(dx\,dy)/dy$ to obtain

$$\begin{aligned} f_{X|Y}(x|y)\,dx &= \frac{f(x, y)\,dx\,dy}{f_Y(y)\,dy} \\ &\approx \frac{P\{x \le X \le x + dx, y \le Y \le y + dy\}}{P\{y \le Y \le y + dy\}} \\ &= P\{x \le X \le x + dx|y \le Y \le y + dy\} \end{aligned}$$

In other words, for small values of dx and dy, $f_{X|Y}(x|y)dx$ represents the conditional probability that X is between x and $x + dx$ given that Y is between y and $y + dy$.

The use of conditional densities allows us to define conditional probabilities of events associated with one random variable when we are given the value of a second random variable. That is, if X and Y are jointly continuous, then, for any set A,

$$P\{X \in A|Y = y\} = \int_A f_{X|Y}(x|y)\,dx$$

In particular, by letting $A = (-\infty, a]$, we can define the conditional cumulative distribution function of X given that $Y = y$ by

$$F_{X|Y}(a|y) \equiv P\{X \le a|Y = y\} = \int_{-\infty}^{a} f_{X|Y}(x|y)\,dx$$

The reader should note that, by using the ideas presented in the preceding discussion, we have been able to give workable expressions for conditional probabilities, even though the event on which we are conditioning (namely, the event $\{Y = y\}$) has probability 0.

EXAMPLE 5a

The joint density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{12}{5}x(2 - x - y) & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Compute the conditional density of X given that $Y = y$, where $0 < y < 1$.

Solution. For $0 < x < 1, 0 < y < 1$, we have

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x,y)}{f_Y(y)} \\ &= \frac{f(x,y)}{\int_{-\infty}^{\infty} f(x,y)\,dx} \\ &= \frac{x(2-x-y)}{\int_0^1 x(2-x-y)\,dx} \\ &= \frac{x(2-x-y)}{\frac{2}{3}-y/2} \\ &= \frac{6x(2-x-y)}{4-3y} \end{aligned}$$

■

EXAMPLE 5b

Suppose that the joint density of X and Y is given by

$$f(x,y) = \begin{cases} \dfrac{e^{-x/y}e^{-y}}{y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find $P\{X > 1|Y = y\}$.

Solution. We first obtain the conditional density of X given that $Y = y$.

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x,y)}{f_Y(y)} \\ &= \frac{e^{-x/y}e^{-y}/y}{e^{-y}\int_0^{\infty}(1/y)e^{-x/y}\,dx} \\ &= \frac{1}{y}e^{-x/y} \end{aligned}$$

Hence,

$$\begin{aligned} P\{X > 1|Y = y\} &= \int_1^{\infty} \frac{1}{y}e^{-x/y}\,dx \\ &= -e^{-x/y}\Big|_1^{\infty} \\ &= e^{-1/y} \end{aligned}$$

■

If X and Y are independent continuous random variables, the conditional density of X given that $Y = y$ is just the unconditional density of X. This is so because, in the independent case,

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x)$$

We can also talk about conditional distributions when the random variables are neither jointly continuous nor jointly discrete. For example, suppose that X is a continuous random variable having probability density function f and N is a discrete random variable, and consider the conditional distribution of X given that $N = n$. Then

$$\frac{P\{x < X < x + dx|N = n\}}{dx} = \frac{P\{N = n|x < X < x + dx\}}{P\{N = n\}} \frac{P\{x < X < x + dx\}}{dx}$$

and letting dx approach 0 gives

$$\lim_{dx \to 0} \frac{P\{x < X < x + dx|N = n\}}{dx} = \frac{P\{N = n|X = x\}}{P\{N = n\}} f(x)$$

thus showing that the conditional density of X given that $N = n$ is given by

$$f_{X|N}(x|n) = \frac{P\{N = n|X = x\}}{P\{N = n\}} f(x)$$

EXAMPLE 5c The Bivariate Normal Distribution

One of the most important joint distributions is the bivariate normal distribution. We say that the random variables X, Y have a bivariate normal distribution if, for constants $\mu_x, \mu_y, \sigma_x > 0, \sigma_y > 0, -1 < \rho < 1$, their joint density function is given, for all $-\infty < x, y < \infty$, by

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1 - \rho^2}} \exp\left\{-\frac{1}{2(1 - \rho^2)}\left[\left(\frac{x - \mu_x}{\sigma_x}\right)^2 + \left(\frac{y - \mu_y}{\sigma_y}\right)^2 - 2\rho\frac{(x - \mu_x)(y - \mu_y)}{\sigma_x\sigma_y}\right]\right\}$$

We now determine the conditional density of X given that $Y = y$. In doing so, we will continually collect all factors that do not depend on x and represent them by the constants C_i. The final constant will then be found by using that $\int_{-\infty}^{\infty} f_{X|Y}(x|y)\, dx = 1$. We have

$$\begin{aligned}
f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} \\
&= C_1 f(x, y) \\
&= C_2 \exp\left\{-\frac{1}{2(1 - \rho^2)}\left[\left(\frac{x - \mu_x}{\sigma_x}\right)^2 - 2\rho\frac{x(y - \mu_y)}{\sigma_x\sigma_y}\right]\right\} \\
&= C_3 \exp\left\{-\frac{1}{2\sigma_x^2(1 - \rho^2)}\left[x^2 - 2x\left(\mu_x + \rho\frac{\sigma_x}{\sigma_y}(y - \mu_y)\right)\right]\right\} \\
&= C_4 \exp\left\{-\frac{1}{2\sigma_x^2(1 - \rho^2)}\left[x - \left(\mu_x + \rho\frac{\sigma_x}{\sigma_y}(y - \mu_y)\right)\right]^2\right\}
\end{aligned}$$

Recognizing the preceding equation as a normal density, we can conclude that, given $Y = y$, the random variable X is normally distributed with mean $\mu_x + \rho\frac{\sigma_x}{\sigma_y}(y - \mu_y)$ and variance $\sigma_x^2(1 - \rho^2)$. Also, because the joint density of Y, X is exactly the same as that of X, Y, except that μ_x, σ_x are interchanged with μ_y, σ_y, it similarly follows that

the conditional distribution of Y given $X = x$ is the normal distribution with mean $\mu_y + \rho\frac{\sigma_y}{\sigma_x}(x - \mu_x)$ and variance $\sigma_y^2(1 - \rho^2)$. It follows from these results that the necessary and sufficient condition for the bivariate normal random variables X and Y to be independent is that $\rho = 0$ (a result that also follows directly from their joint density, because it is only when $\rho = 0$ that the joint density factors into two terms, one depending only on x and the other only on y).

With $C = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$, the marginal density of X can be obtained from

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y)\, dy \\ &= C\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1 - \rho^2)}\left[\left(\frac{x - \mu_x}{\sigma_x}\right)^2 + \left(\frac{y - \mu_y}{\sigma_y}\right)^2 \right.\right. \\ &\qquad \left.\left. -2\rho\frac{(x - \mu_x)(y - \mu_y)}{\sigma_x\sigma_y}\right]\right\} dy \end{aligned}$$

Making the change of variables $w = \frac{y-\mu_y}{\sigma_y}$ gives

$$\begin{aligned} f_X(x) &= C\sigma_y \exp\left\{-\frac{1}{2(1 - \rho^2)}\left(\frac{x - \mu_x}{\sigma_x}\right)^2\right\} \\ &\quad \times \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1 - \rho^2)}\left[w^2 - 2\rho\,\frac{x - \mu_x}{\sigma_x}\,w\right]\right\} dw \\ &= C\sigma_y \exp\left\{-\frac{1}{2(1 - \rho^2)}\left(\frac{x - \mu_x}{\sigma_x}\right)^2 (1 - \rho^2)\right\} \\ &\quad \times \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1 - \rho^2)}\left[w - \rho\,\frac{x - \mu_x}{\sigma_x}\right]^2\right\} dw \end{aligned}$$

Because

$$\frac{1}{\sqrt{2\pi(1 - \rho^2)}}\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1 - \rho^2)}\left[w - \frac{\rho}{\sigma_x}(x - \mu_x)\right]^2\right\} dw = 1$$

we see that

$$\begin{aligned} f_X(x) &= C\sigma_y\sqrt{2\pi(1 - \rho^2)}\, e^{-(x-\mu_x)^2/2\sigma_x^2} \\ &= \frac{1}{\sqrt{2\pi}\sigma_x}\, e^{-(x-\mu_x)^2/2\sigma_x^2} \end{aligned}$$

That is, X is normal with mean μ_x and variance σ_x^2. Similarly, Y is normal with mean μ_y and variance σ_y^2. ■

EXAMPLE 5d

Consider $n + m$ trials having a common probability of success. Suppose, however, that this success probability is not fixed in advance but is chosen from a uniform $(0, 1)$

population. What is the conditional distribution of the success probability given that the $n + m$ trials result in n successes?

Solution. If we let X denote the probability that a given trial is a success, then X is a uniform $(0, 1)$ random variable. Also, given that $X = x$, the $n + m$ trials are independent with common probability of success x, so N, the number of successes, is a binomial random variable with parameters $(n + m, x)$. Hence, the conditional density of X given that $N = n$ is

$$\begin{aligned} f_{X|N}(x|n) &= \frac{P\{N = n|X = x\}f_X(x)}{P\{N = n\}} \\ &= \frac{\binom{n+m}{n} x^n(1-x)^m}{P\{N = n\}} \qquad 0 < x < 1 \\ &= cx^n(1-x)^m \end{aligned}$$

where c does not depend on x. Thus, the conditional density is that of a beta random variable with parameters $n + 1, m + 1$.

The preceding result is quite interesting, for it states that if the original or *prior* (to the collection of data) distribution of a trial success probability is uniformly distributed over $(0, 1)$ [or, equivalently, is beta with parameters $(1, 1)$] then the posterior (or conditional) distribution given a total of n successes in $n + m$ trials is beta with parameters $(1 + n, 1 + m)$. This is valuable, for it enhances our intuition as to what it means to assume that a random variable has a beta distribution. ■

*6.6 ORDER STATISTICS

Let $X_1, X_2, \ldots, X_n$ be n independent and identically distributed continuous random variables having a common density f and distribution function F. Define

$$\begin{aligned} X_{(1)} &= \text{smallest of } X_1, X_2, \ldots, X_n \\ X_{(2)} &= \text{second smallest of } X_1, X_2, \ldots, X_n \\ &\vdots \\ X_{(j)} &= j\text{th smallest of } X_1, X_2, \ldots, X_n \\ &\vdots \\ X_{(n)} &= \text{largest of } X_1, X_2, \ldots, X_n \end{aligned}$$

The ordered values $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ are known as the *order statistics* corresponding to the random variables $X_1, X_2, \ldots, X_n$. In other words, $X_{(1)}, \ldots, X_{(n)}$ are the ordered values of $X_1, \ldots, X_n$.

The joint density function of the order statistics is obtained by noting that the order statistics $X_{(1)}, \ldots, X_{(n)}$ will take on the values $x_1 \le x_2 \le \cdots \le x_n$ if and only if, for some permutation $(i_1, i_2, \ldots, i_n)$ of $(1, 2, \ldots, n)$,

$$X_1 = x_{i_1}, X_2 = x_{i_2}, \ldots, X_n = x_{i_n}$$